Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption

One potential explanation for the website's connection timeout error message is: The web server may have experienced a Denial of Service (DoS) attack. The attacker sent too many connection requests to the server, making it difficult for the website to respond to normal users.

The logs show that: A large number of TCP SYN packets were coming from an unfamiliar IP address. The server was receiving more connection requests than it could handle, causing the website to become slow and eventually stop responding.

This event could be:A TCP SYN flood attack, which is a type of Denial of Service (DoS) attack. The attacker sends many SYN requests without completing the connection process, causing the server to use up resources while waiting for responses.

Section 2: Explain how the attack is causing the website to malfunction

When website visitors try to establish a connection with the web server, a three-way handshake occurs using the TCP protocol. Explain the three steps of the handshake:
1.SYN: The user's device sends a SYN packet to the web server to request a connection.

2. SYN-ACK: The server responds with a SYN-ACK packet to acknowledge the request and indicate that it is ready to connect.

3. ACK: The user's device sends an ACK packet back to the server to complete the connection..

Explain what happens when a malicious actor sends a large number of SYN packets all at once: When an attacker sends too many SYN packets at once, the server tries to respond to each request. However, the attacker does not complete the three-way handshake. This leaves many connections half-open, using up the server's resources and making it difficult for legitimate users to connect.

Explain what the logs indicate and how that affects the server: The packet capture shows an unusually large number of TCP SYN requests from an unfamiliar IP address. This suggests that the server is experiencing a SYN flood attack.Because the server is overwhelmed, it cannot respond properly to normal website requests. This causes slow loading times and connection timeout errors.

As a result, employees cannot access the company's website to search for vacation packages, which can affect customer service and business operations. To prevent similar attacks, the company can use firewall rules, SYN flood protection, and network monitoring tools to detect and limit suspicious traffic.